

GRADE: Generalization Robustness Assessment for AI-Text Detectors in Education — Do Detectors Survive a Change of Cohort?

Abstract

Universities are adopting AI-text detectors as academic-integrity controls on the strength of reported accuracies above 99%. Those figures are obtained on the same student work a detector learned from, which is not the situation a programme faces when screening next semester's assignment. We present GRADE (Generalization Robustness Assessment for Detectors in Education), a protocol that trains five detectors on one collection of student essays and tests them on writing withheld entirely from training, and report early findings from applying it. On one widely used collection, a classifier given nothing but the length of an essay reaches 80.3% accuracy: a detector could report a strong figure there without recognising authorship at all. On our own collection that comparison falls to 54.5%, so whether length passes for authorship depends on how the evaluation set was assembled. We set out what evidence a university should require before placing such a tool inside an integrity process.

Keywords: academic integrity, assessment, generative AI, AI-text detection, educational technology adoption, evaluation methodology.

1. Introduction

Generative language models have unsettled written assessment. A student can now produce a competent essay without doing the thinking the essay was set to evidence, and institutions are under pressure to respond. One response has been procedural, through redesigned assessment. Another has been technical: adopt a detector, run submissions through it, and treat the output as evidence in academic-integrity processes.

The technical response is being justified by numbers. Recent work reports Macro F1 scores of 99.78% and 98.10% for fine-tuned detectors, rising to 100.00% on one corpus [1]. Vendors of commercial tools make comparable claims. Read at face value, these figures suggest the problem is solved.

They are measured under conditions no institution reproduces. A detector is trained and evaluated on the same corpus, so the essays it is scored against answer the same prompts, come from the same cohort, and were generated by the same models as the essays it learned from. What a program director needs to know is different: what happens with next semester's assignment, next year's intake, and the model released after the detector was built.

This gap matters to Information Systems education specifically, because adopting a detector is an information-systems decision. It places an automated classifier inside a consequential process about students, with sanctions attached to its output. IS scholarship has long held that evaluating a system means evaluating it under its conditions of use, and that evidence generated under laboratory conditions does not transfer automatically to deployment. Detection tools are being adopted without that step.

The paper therefore speaks to academic integrity and the role of AI in assessment environments: it examines an intelligent system deployed by educators to govern assessment, rather than the generative systems students use. It also bears on AI-resilient assessment design, since evidence that detection transfers poorly across cohorts is an argument for assessment that does not depend on it.

We are conducting a study to supply the missing evidence. It asks a deliberately narrow question: *if an encoder detector is trained on one corpus of student essays and then meets essays from a different*

source, how much of its reported performance survives? This short paper sets out the motivation, the research design, and the contributions we expect, together with two findings already established: that a measurement of essay length alone can pass for detection on one widely used collection, and that whether it can depends on how that collection was assembled.

2. Background and Related Work

2.1. What detection studies report

Supervised transformer detectors dominate current practice. Qorich and El Ouazzani fine-tune GPT-2, RoBERTa and BART across five corpora and report the figures quoted above [1]. Their strongest result, 100.00% accuracy and Macro F1, is obtained on a corpus in which every AI text is a paraphrase of the human text it is paired with, under a split that is stratified but not grouped by source text. A paraphrase can therefore be scored in the test set while its own source sits in training. A perfect score on 100,000 items is more readily explained by that arrangement than by detection.

This is not an isolated lapse. In prior work we measured that under a row-level split, 46 to 53 per cent of test items had their source document present in training, allowing a detector to succeed partly by recognising a familiar topic rather than a synthetic style [2]. Grouping the split by source identifier removes the leak and lowers the reported numbers. The lower numbers are the honest ones.

2.2. What education-domain studies report

Work situated in education reaches conclusions difficult to reconcile with near-perfect accuracy. Weber-Wulff et al. evaluate fourteen tools and find them unreliable for academic use, particularly under obfuscation [6]. Liang et al. show detectors are biased against non-native English writers [7], which in an assessment context means the cost of false accusation falls unevenly across a cohort. Sadasivan et al. question whether reliable detection is achievable in principle [8].

Two literatures thus coexist: one reporting that detection is essentially solved, another that it fails where it would be used. We take the discrepancy to be partly a measurement artefact. The first literature reports in-distribution performance on corpora with exploitable structure; the second reports deployment performance. The contribution we aim at is to measure the distance between them in the education domain, under conditions stated explicitly enough to be checked.

2.3. Distribution shift

Robustness under shift is established as a concern. MAGE reports substantial degradation on unseen generators [9], and paraphrasing is a known evasion strategy [10]. Our own prior work rotated the held-out generator and found that two detectors with statistically indistinguishable in-distribution accuracy, 97.70% and 97.77%, diverged sharply once a generator was withheld: recall fell by 18.79 points for one and 1.36 for the other [2]. That study concerned generator shift on web text. The present study concerns corpus shift on student essays, which is the shift an institution meets and which has received less attention.

3. Research Motivation

An institution deciding whether to adopt a detector is making a procurement decision under uncertainty, and the evidence available is mismatched to it in three ways.

The evidence is in-distribution. A reported accuracy describes performance on essays resembling the training data. Deployment presents a different assignment, a different cohort, and generators that postdate the detector. Nothing in the reported figure speaks to this.

The reported metric is the wrong one. Accuracy weights both errors equally. In assessment they are not equal and they are not symmetric. A false negative returns an AI-written essay as authentic: the

integrity control silently fails. A false positive accuses a student who wrote their own work, triggering a process with real consequences and, per Liang et al., doing so unevenly across the cohort [7]. The two errors also fall on different parties: a false negative costs the institution its control, while a false positive costs an individual student the presumption of good faith. An institution needs recall, false-negative rate and false-positive rate separately, and it needs them on data resembling its own. Accuracy conceals the trade-off it most needs to see.

The evidence lacks controls. A corpus can separate its classes through properties unrelated to writing. If human essays in an evaluation corpus are systematically longer than AI ones, a classifier can score well by measuring length. Detection papers seldom report what a trivial baseline achieves on the same data, so a reader cannot tell how much of a result reflects authorship. We show below that this is not hypothetical.

These three concerns motivate our design: evaluate under shift, report the error types that matter for assessment, and publish control baselines alongside detector performance.

4. Research Design

The principle of **GRADE** is that a detector is characterised not by a single accuracy figure but by the distance between what it scores on the student work it was built from and what it scores on writing withheld entirely from training and model selection. That distance, rather than the headline number, is what an institution is buying.

4.1. The detectors under test

We build five detectors, each from a different pretrained language model, and train each to answer one question about a submitted essay: was this written by a student or generated. The five models are `bert-base-uncased`, `roberta-base`, `deberta-v3-small`, `distilbert-base-uncased` and `electra-base-discriminator`. They are of comparable size and receive an identical training procedure (Table 2), so any difference in behaviour reflects the model rather than how much effort we spent tuning it.

We test five rather than one because an institution selecting a tool is choosing among alternatives, often on the basis of a single headline accuracy. If those five look equivalent on the essays they were built from but diverge on essays from elsewhere, the headline figure is not a usable basis for that choice. Detectors of this kind are the technology underneath the commercial tools now marketed to universities, so their behaviour is of direct practical interest to anyone weighing adoption.

4.2. Student work and academic writing used

Table 1 summarises the three collections. We arrived at these after examining six: three others were discarded because their labels could not be trusted, in one case because the word [AI-Generated] was printed at the start of every generated document, and in another because both classes were assembled from the same small pool of sentences. Collections offered for this purpose are not uniformly sound, and we checked before using rather than after.

What the detectors learn from. We train on the LLM-Detect AI Generated Text collection from the competition organised by Vanderbilt University and The Learning Agency Lab [3]. These are genuine student essays, with student-level spelling and grammar intact, answering one of two assigned prompts: an explanatory essay on limiting car usage, and a letter arguing for or against the Electoral College. Its AI half was generated with PaLM 2, Gemini 1.0 Pro, Gemini 1.0 Ultra and GPT-4, three-shot prompted on student essays of the same topic, deduplicated at an 85% embedding-similarity threshold, then distribution-matched to the student essays on word count, sentence length and function-word count. The AI essays were therefore built to resemble the human ones, which makes this corpus harder, and closer to the classroom case, than one assembled from unmatched sources.

Table 1: Student work and academic writing used. The first collection is divided for training, stopping and marking; the other two are opened once, at the end.

Collection	Role	Items	Student:AI
LLM-Detect [3] student essays, 2 prompts	train / stop / mark	27,222	16,042 : 11,180
AI Text Pile [4] web and chatbot text	unseen	19,214	9,607 : 9,607
Abstract Para. [5] PhD abstracts, GPT-4	unseen	304	152 : 152

Table 2: Training procedure, identical for all five detectors so that any difference in behaviour reflects the model rather than the effort spent tuning it.

Setting	Value
Essay length considered	first 256 tokens
Optimiser	AdamW
Learning rate	2×10^{-5}
Batch size	8
Passes over the data	5
Loss	cross-entropy
Parameters updated	all (model and classifier)
Model retained	best on the stopping portion
Training runs	1 (seed 42)

What the detectors are then tested on. We evaluate on two collections absent from training and from model selection: a balanced 20,000-item sample of the AI Text Detection Pile [4], MIT licensed, whose human side is drawn from web and essay sources and whose AI side comes from GPT-2, GPT-3, GPT-J and ChatGPT; and the Abstract Paraphrasing corpus [5], 304 English PhD-thesis abstracts, half paraphrased by GPT-4 at temperature zero. The first differs from the training corpus in provenance, genre and generator. The second supplies a formal academic register and a generator the training data does not contain, but at 304 items it is underpowered and we report it as a direction rather than a result.

4.3. Guarding against a flattering evaluation

The essays we train on are divided three ways: 60% to teach the detector, 20% to decide when to stop teaching it, and 20% held back to mark it. The two external collections are opened once, at the very end. Nothing about them influences how the detectors are built, which is what makes a result on them meaningful.

The obvious pitfall is marking a detector on work it has already seen — the equivalent of setting an exam from the worked examples. We therefore check rather than assume: an automated audit confirms that no essay appears on both sides of the division, and it halts the study rather than issuing a warning if one does. Each portion is also evened out so that half the essays are student-written and half generated, which fixes the score of a detector that guesses at 50% and makes the reported figures directly comparable across collections.

We report accuracy, precision, recall, F1, area under the ROC curve, and the two error rates separately: the false-positive rate, being the share of genuine student essays wrongly flagged, and the false-negative rate, being the share of generated essays passed as authentic. The second is emphasised throughout for the reason given in Section 3.

Alongside every detector we run two deliberately unintelligent comparisons on each collection. The first is given only one piece of information about each essay: how long it is. The second sees only which words appear, with no sense of their order or meaning. Neither can be said to recognise authorship. They exist to establish the score a detector must beat before its performance can be attributed to anything about writing, and a detector that fails to clear them on a given collection will be reported as having demonstrated nothing there.

5. Preliminary Findings

Training is in progress at the time of submission. Two findings are already established, and both bear on how a reported accuracy should be read.

5.1. An essay's length can pass for its authorship

On the AI Text Detection Pile, the median human document is 7,332 characters and the median AI document 1,260, a ratio of roughly six to one. A logistic classifier whose only feature is document length reaches **80.3%** accuracy there; a TF-IDF classifier reaches 97.4%. A detector reporting 85% on this corpus would have added little to a measurement of length.

The comparison with our training corpus is instructive. There the median human essay is 2,180 characters and the median AI essay 2,003, and the length-only control falls to **54.5%** against a 50% chance baseline. The distribution-matching applied when that corpus was built has evidently worked, and it shows that the confound is a property of how a corpus was assembled rather than an inevitable feature of the task. This is precisely why the control must be reported per corpus.

We mitigate the effect by truncating to 256 tokens, which removes part of it since both classes exceed 256 tokens at the median, and quantify the remainder by reporting the length-only control beside every detector result. Note also that TF-IDF reaches 98.6% on our training corpus in distribution: a fine-tuned transformer scoring in that range would not have improved on bag-of-words, which is itself a reason to report shift performance rather than in-distribution accuracy alone.

5.2. Detector results

Training is under way at the time of submission. Tables 3 and 4 give the reporting format we will use, with the collections and comparisons fixed and the measured cells left empty. We show them here so that the basis of the eventual claim is visible in advance: the argument of this paper is that evaluation conditions should be declared before results arrive, and we hold ourselves to that.

Table 3 places every detector beside the two unintelligent comparisons on the same collection, so a reader can see at once whether a given figure represents recognition of writing or a property of the collection. Table 4 reduces each detector to the quantity an institution actually needs: how far its performance falls when the essays come from somewhere else. We will report the direction of these results once measured, and not before.

The final row of Table 4 is already informative. The length-only comparison *improves* by 25.8 points when it moves to the unseen collection, because that collection's generated items are shorter than its student writing. A detector could post a higher figure there while having become less able to recognise authorship, which is the clearest illustration we can offer of why a single accuracy figure should not settle an adoption decision.

6. Expected Contributions

To assessment practice. A quantified statement of what happens to detector performance when the corpus changes, expressed in the terms an adoption decision needs: how much recall is lost, and therefore what share of AI-written work a detector can be expected to return as authentic outside its training distribution.

Table 3: Reporting format for detector performance (**measured values pending**). Generated text is the positive class; all figures are percentages. Each block gives the five detectors followed by the two unintelligent comparisons on the same collection. Single training run, seed 42.

Evaluation collection	Detector	Acc.	Prec.	Recall	ROC	FPR	FNR
<i>Same collection as training</i> (4,462 essays, balanced)							
	BERT	—	—	—	—	—	—
	RoBERTa	—	—	—	—	—	—
	DeBERTa-v3	—	—	—	—	—	—
	DistilBERT	—	—	—	—	—	—
	ELECTRA	—	—	—	—	—	—
	<i>word-frequency comparison</i>	98.6	—	97.9	99.9	—	—
	<i>length-only comparison</i>	54.5	—	56.5	60.5	—	—
<i>Unseen collection: AI Text Pile</i> (19,214 items, balanced)							
	BERT	—	—	—	—	—	—
	RoBERTa	—	—	—	—	—	—
	DeBERTa-v3	—	—	—	—	—	—
	DistilBERT	—	—	—	—	—	—
	ELECTRA	—	—	—	—	—	—
	<i>word-frequency comparison</i>	97.4	—	96.6	99.7	—	—
	<i>length-only comparison</i>	80.3	—	79.3	93.0	—	—
<i>Unseen collection: academic abstracts</i> (304 items, balanced; underpowered)							
	BERT	—	—	—	—	—	—
	RoBERTa	—	—	—	—	—	—
	DeBERTa-v3	—	—	—	—	—	—
	DistilBERT	—	—	—	—	—	—
	ELECTRA	—	—	—	—	—	—
	<i>word-frequency comparison</i>	80.3	—	73.7	88.4	—	—
	<i>length-only comparison</i>	55.3	—	73.7	59.3	—	—

To how these tools are evaluated in education. A protocol, GRADE, that combines an enforced check against marking a detector on work it has already seen with a requirement to report what an unintelligent comparison achieves on the same essays. Our length finding shows this is not a formality: on one collection in current use, that comparison alone reaches 80.3%.

To institutional decision-making. On the evidence so far we propose that a university evaluating a detector should require, at minimum: performance on student work the supplier did not build the tool from; the two error rates reported separately rather than a single accuracy; evidence that the result is not an artefact of how the evaluation set was assembled; and a stated position on whether error rates differ across groups of students, given the documented bias against non-native writers [7]. We expect the completed study to let us attach numbers to the first three.

To AI-resilient assessment design. We do not claim detection is the right response to generative AI in assessment, and evidence that detectors transfer poorly across corpora is also evidence for the alternative. An institution that cannot rely on a detector outside the distribution it was built on has a stronger case for assessment designs that do not depend on authorship inference at all: process evidence, oral defence, or work that is situated enough to be difficult to outsource. Our contribution to that argument is empirical rather than pedagogical, and we regard it as the more durable direction.

7. Limitations and Next Steps

We state these plainly, since the argument of this paper is that evaluation conditions should be reported honestly.

Two prompts. Our training corpus covers two assigned prompts, so it supports a claim about corpus

Table 4: Reporting format for the shortfall under a change of collection (**measured values pending**). The recall column is given separately because a generated essay passed as authentic is the failure an integrity control is adopted to prevent.

Detector	Accuracy		Recall	
	Same	Shortfall	Same	Shortfall
BERT	—	—	—	—
RoBERTa	—	—	—	—
DeBERTa-v3	—	—	—	—
DistilBERT	—	—	—	—
ELECTRA	—	—	—	—
<i>length-only comparison</i>	54.5	−25.8	56.5	−22.7

shift but only a weak one about topic shift, and none about the range of assignments institutions set.

Single seed. Each configuration is trained once. We report no variance and will not claim that small differences between architectures are reliable. In prior work, accuracy varied by ± 6.35 points and recall by ± 13.70 points across three seeds on a hard held-out fold [2], so differences of that order here must be read as provisional.

No cross-generator claim. The training corpus carries a binary label only. We cannot say which generators are harder to detect.

Underpowered probe. At 304 items, one misclassification moves accuracy on the Abstract Paraphrasing corpus by roughly 0.7 points.

No pedagogical claim. Section 6 argues that poor transfer strengthens the case for AI-resilient assessment, but we study detectors, not assessment designs. We offer no evidence about which alternative forms of evaluation work.

Next steps are to complete the training runs and report the detectors against the comparisons in the format of Tables 3 and 4, then to extend the evaluation to student writing from a wider range of assignments than the two represented here.

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